

The Definite Integral (5.2)

Suppose we have a function $f(x)$ defined on an interval $[a, b]$, and $f(x)$ is positive on this interval. We can approximate the area under the graph of $f(x)$ by subdividing $[a, b]$ into smaller intervals of length Δx choosing a point x_i^* in each interval, and summing up $f(x_i^*)\Delta x$ over all x_i^* .



Consider a small interval $[x_i, x_{i+1}]$

Some possible choices of x_i^* :

- Left endpoint of interval
- Right endpoint of interval
- Midpoint of interval

If we subdivide into smaller and smaller intervals, we get a better approximation. The true area is given by

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x$$

If f takes on both positive and negative values, then we can still subdivide $[a, b]$ into intervals of length Δx , choose x_i^* from each interval, and compute

$$\sum_{i=1}^n f(x_i^*) \Delta x$$

Negative values of $f(x_i^*)$ will contribute negatively to the sum.

Definition: Let f be a function defined on an interval $[a, b]$. The **definite integral** of f from a to b is

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x$$

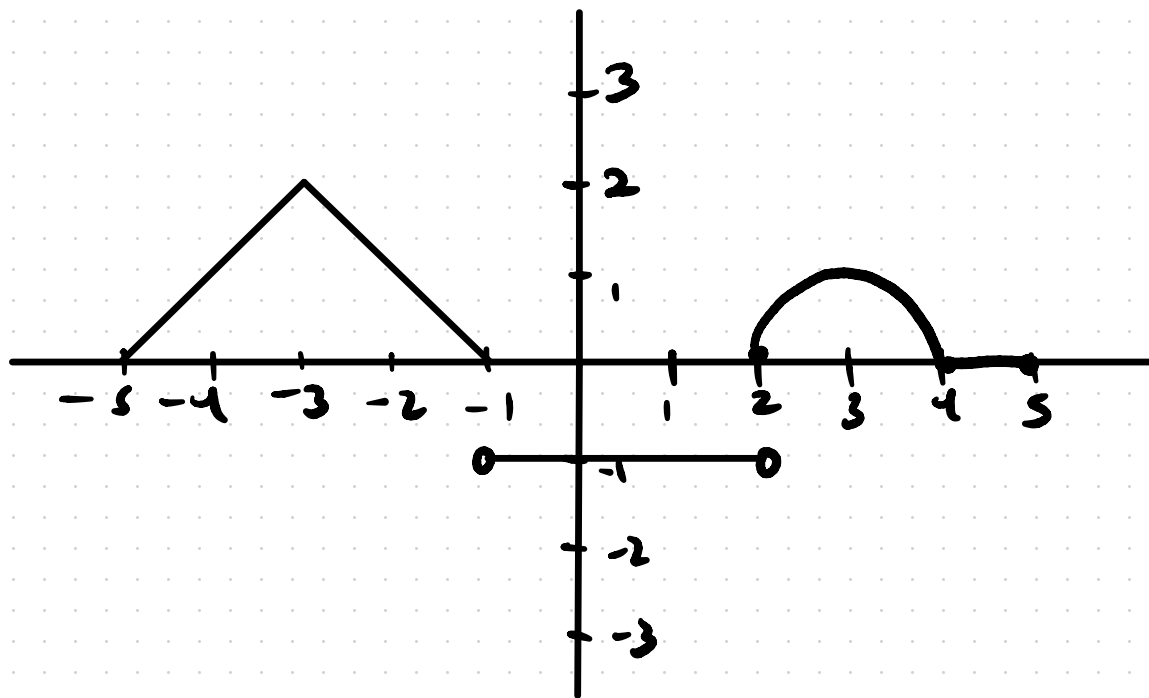
where $\Delta x = \frac{b-a}{n}$ (length of a small interval), and x_i^* is chosen from the i^{th} interval. We denote the definite integral by

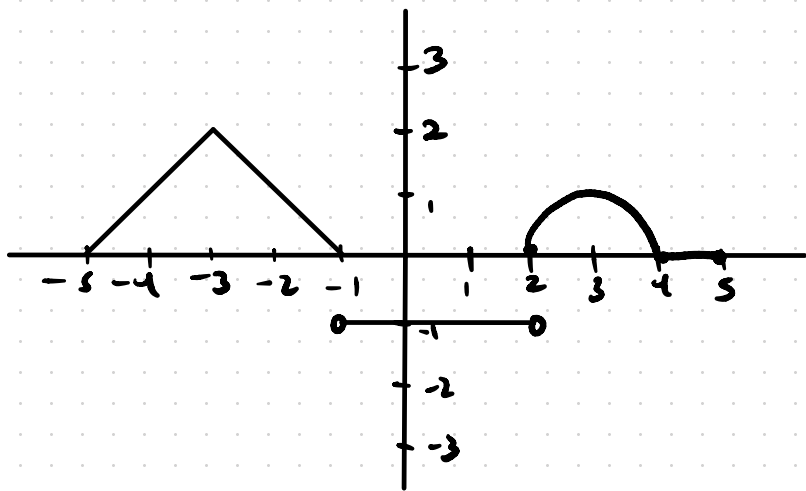
Graphically,

$$\int_a^b f(x) dx$$

is the

Let f be the following function.





$$\int_{-5}^{-1} f(x) dx$$

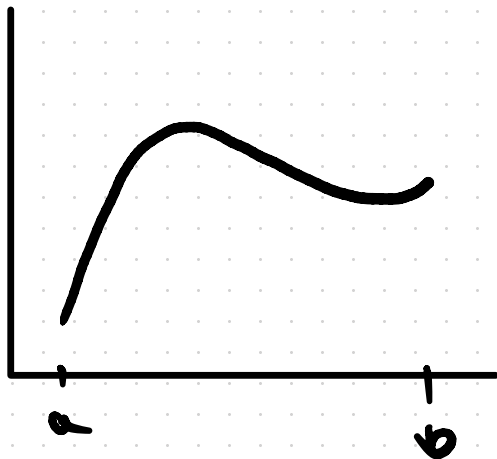
$$\int_{-5}^1 f(x) dx$$

$$\int_{-1}^4 f(x) dx$$

$$\int_4^5 f(x) dx$$

Why the funny notation?

$$\int_a^b f(x) dx$$



Evaluate

$$\int_{-3}^3 |2x-1| dx$$

Evaluate

$$\int_3^7 dx$$

Evaluate

$$\int_{10}^{10} \frac{x^3 \sin^4 x}{\sqrt{\ln x + \tan x}} dx$$

Basic properties of definite integrals

$$\int_a^b (f(x) + g(x)) dx = \int_a^b f(x) dx + \int_a^b g(x) dx$$

$$\int_a^b (f(x) - g(x)) dx = \int_a^b f(x) dx - \int_a^b g(x) dx$$

$$\int_a^b c f(x) dx = c \int_a^b f(x) dx$$

c a constant

$$\int_a^b f(x)dx + \int_b^c f(x)dx = \int_a^c f(x)dx$$

If $f(x) \leq g(x)$ for all $a \leq x \leq b$, then

$$\int_a^b f(x) dx \leq \int_a^b g(x) dx$$